TransUNet-Based Brain Tumor Segmentation with Graph Cut Refinement and Non-Clinical Decision Support

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***Abstract*—Brain tumor segmentation from multi-modal magnetic resonance imaging (MRI) is essential for accurate diagnosis and pre-surgical planning. However, it remains challenging due to irregular tumor shapes, heterogeneous intensity distribution, and unclear boundaries. In this work, we propose a hybrid deep learning framework that combines a U-Net architecture with Transformer modules to capture both local features and long-range spatial dependencies. The proposed model is trained and evaluated on the BraTS 2020 dataset, using multi-modal MRI sequences (T1, T1ce, T2, and FLAIR) to ensure robust feature representation. To further improve segmentation quality, a graph cut–based refinement method is applied as a post-processing step to enhance boundary accuracy and reduce noise in the predicted masks. In addition to segmentation, we introduce a Non-Clinical Decision Support System (NCDSS) to provide basic tumor analysis. First, a volumetric estimation is performed by calculating the tumor size based on pixel distribution across 3D MRI slices. Second, an atlas-based localization method is used to map the segmented tumor onto standard anatomical regions. This enables identification of whether the tumor is located in critical brain areas thereby indicating if the tumor lies in a potentially high-risk region. Experimental results show that the proposed method achieves a Dice Similarity Coefficient (DSC)**

***Index Terms*—Brain tumor segmentation, U-Net, Transformers, Graph Cut, Atlas-based Localization, BraTS 2020, MRI, Decision Support System, Volumetric Analysis, Eloquent Regions**

I. Introduction

Brain tumors represent one of the most life-threatening neurological disorders, characterized by abnormal and uncontrolled growth of cells within brain tissues. Early detection and precise characterization of these tumors are crucial for effective treatment planning, surgical intervention, and patient survival. Magnetic Resonance Imaging (MRI) is the most widely used imaging modality for brain tumor diagnosis due to its superior soft tissue contrast and non-invasive nature. However, extracting meaningful information from MRI scans, such as tumor boundaries, location, and volume, remains a challenging and labor-intensive task when performed manually by radiologists.

Manual segmentation suffers from several limitations, including high dependency on expert knowledge, time consumption, and variability across different observers. These challenges have led to a growing interest in automated brain tumor segmentation systems. In recent years, deep learning techniques have demonstrated remarkable success in medical image analysis, particularly in segmentation tasks. Among these, the U-Net architecture has emerged as a dominant approach due to its ability to effectively capture fine-grained spatial details through skip connections between encoder and decoder layers. Despite its strengths, conventional U-Net models primarily focus on local features and often struggle to capture global contextual information, which is essential for accurately identifying irregular and diffuse tumor regions.

To overcome this limitation, transformer-based architectures have been introduced into computer vision tasks. Transformers, originally designed for natural language processing, rely on self-attention mechanisms to model long-range dependencies and global relationships within data. Integrating transformers into convolutional architectures enables a more comprehensive understanding of both local and global features. In this research, we enhance the traditional U-Net by incorporating a transformer module at the bottleneck layer. This hybrid design allows the model to retain the spatial precision of convolutional layers while leveraging the contextual awareness provided by transformers, leading to improved segmentation performance in complex tumor structures.

While deep learning models can produce high-quality segmentation outputs, they may still generate noisy predictions or imprecise boundaries due to variations in image intensity, tumor heterogeneity, and limited training data. To address these issues, post-processing techniques are essential. In this work, we employ graph cut–based refinement, a classical optimization approach that formulates segmentation as an energy minimization problem. Graph cut methods enforce spatial consistency and boundary smoothness, effectively reducing false positives and improving the delineation of tumor margins. This step enhances the robustness and reliability of the segmentation results, making them more suitable for practical applications.

Beyond segmentation, understanding the anatomical context of the tumor is equally important. Identifying the exact location of a tumor within the brain can provide critical insights into its potential impact on cognitive and motor functions. To achieve this, our framework integrates the Harvard brain atlas, which provides a standardized mapping of brain regions. By aligning the segmented tumor with the atlas, we can determine the specific anatomical regions affected by the tumor. Furthermore, quantitative analysis is performed to estimate tumor size and volume, which are key indicators in clinical assessment.

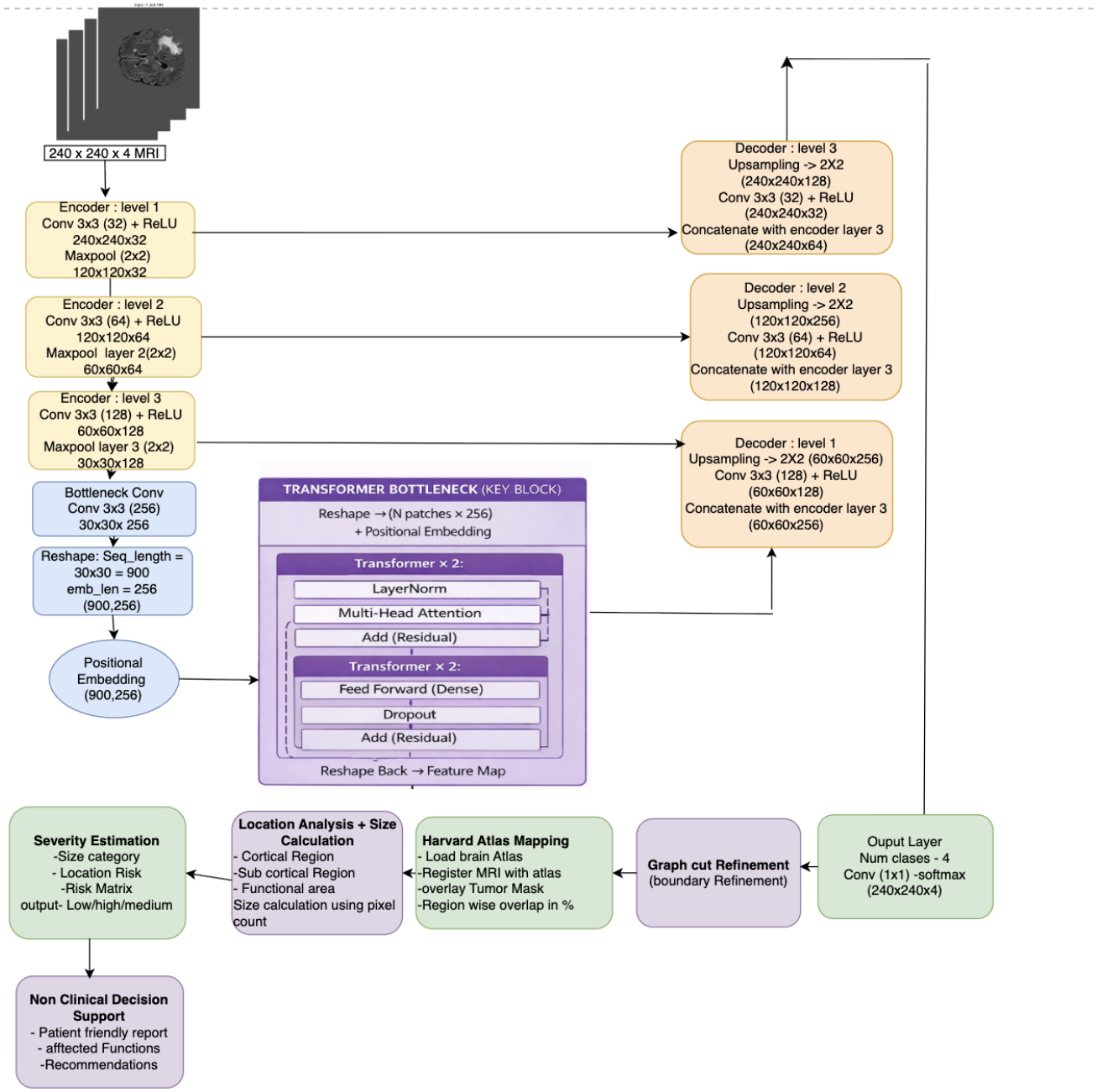


Fig. 1: Proposed Brain Tumor Segmentation Architecture

Building upon these analyses, the proposed system introduces a non-clinical decision support component. This module evaluates tumor characteristics such as size, location, and affected regions to estimate severity levels. The goal is not to replace medical professionals but to assist in preliminary assessment, prioritization, and improved understanding of tumor conditions. Such systems can be particularly beneficial in resource-constrained environments where access to expert radiologists may be limited.

The integration of deep learning, transformer-based contextual modeling, graph-based optimization, and anatomical mapping distinguishes this work from conventional segmentation approaches. By combining these methodologies, the proposed framework aims to deliver a comprehensive and interpretable solution for brain tumor analysis. This research contributes toward bridging the gap between automated image segmentation and intelligent decision support systems, ultimately supporting advancements in medical imaging and healthcare technology. The ultimate objective of this study is to encapsulate these predictive insights into a comprehensive non-clinical Decision Support System (DSS). Designed to operate safely outside the bounds of primary diagnostic medical devices, this DSS serves as an advanced informational tool and facilitating multidisciplinary team discussions.

II. MRI Segmentation Algorithm of Brain Tumor Based on U-Net with Transformer Bottleneck and Graph Cut Refinement

A. Network Structure

The proposed framework adopts a hybrid deep learning architecture designed to effectively segment brain tumors from MRI scans by combining convolutional feature extraction, transformer-based contextual modeling, and optimization-based refinement. The network follows an encoder–bottleneck–decoder paradigm, where the encoder extracts hierarchical spatial features, the bottleneck enhances global contextual understanding, and the decoder reconstructs the segmentation mask.

Given an input MRI slice $X \in R^{H \times W \times C}$, the encoder maps it to a latent representation F_e, such that:

$$F_e = \mathcal{E}(X) \tag{1}$$

where $\mathcal{E}(\cdot)$ denotes the encoding function composed of stacked convolutional layers and pooling operations. This representation is then processed by a transformer module at the bottleneck, producing a context-enhanced feature map F_t:

$$F_t = \mathcal{T}(F_e) \tag{2}$$

The decoder reconstructs the segmentation output $\hat{Y}$ from the transformed features:

$$\hat{Y} = \mathcal{D}(F_t, S) \tag{3}$$

where S represents skip connections from the encoder that preserve spatial details. Finally, a graph cut refinement stage is applied to obtain the final segmentation Y^*, ensuring spatial consistency:

$$Y^* = \mathcal{G}(\hat{Y}) \tag{4}$$

This structured pipeline allows the model to integrate multi-level information, combining local spatial precision with global contextual awareness and post-processing optimization.

B. TransUNet Architecture with Transformer Bottleneck

The TransUNet architecture enhances the conventional U-Net by embedding a transformer module at the bottleneck layer. The encoder consists of convolutional blocks that extract local features using operations defined as:

$$F_{l+1} = \sigma(W_l * F_l + b_l) \tag{5}$$

where $*$ denotes convolution, W_l and b_l are learnable parameters, and σ is a non-linear activation function.

At the bottleneck, the feature map $F_e \in R^{h \times w \times d}$ is reshaped into a sequence of flattened patches:

$$Z_0 = [x_1, x_2, \ldots, x_N], \quad x_i \in R^d \tag{6}$$

Positional embeddings P are added to retain spatial information:

$$Z_0 = Z_0 + P \tag{7}$$

The sequence is then processed through transformer encoder layers using self-attention mechanisms:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{8}$$

where Q, K, and V are projections of the input sequence. This mechanism enables the model to capture long-range dependencies across the entire image. The output is reshaped back into a feature map and passed to the decoder, which reconstructs the segmentation mask using upsampling and skip connections.

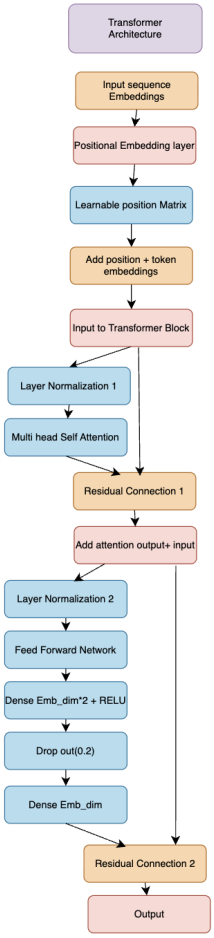


Fig. 2: Transformer Architecture

C. Scale-Aware Enhancement Module

To address the variability in tumor size and morphology, a Scale-Aware Enhancement Module (SAEM) is incorporated to capture multi-scale contextual information. This module processes feature maps using multiple convolutional filters with different receptive fields, generating a set of scale-specific features $\{F_1, F_2, \ldots, F_n\}$. These features are then fused using learnable weights:

$$F_{\text{saem}} = \sum_{i=1}^{n} \alpha_i F_i \tag{9}$$

where α_i represents the importance of each scale and satisfies:

$$\sum_{i=1}^{n} \alpha_i = 1 \tag{10}$$

This formulation enables the network to adaptively emphasize relevant scales, improving segmentation performance for both small and large tumor regions. The enhanced feature map is subsequently integrated into the decoder to refine the final prediction.

D. Graph Cut-Based Refinement

The graph cut refinement stage enhances the initial segmentation obtained from the deep learning model by enforcing spatial consistency and boundary smoothness through an energy minimization framework. The probabilistic output of the segmentation network, denoted as $\hat{Y}$, is interpreted as a soft labeling of each pixel, where each pixel is associated with class probabilities. These probabilities are converted into unary energy terms that guide the graph cut optimization.The goal is to find the most probable labeling configuration that explains the observed image data.

Let the predicted probability map be represented as $P = \{p_0, p_1, \ldots, p_K\}$, where p_0 corresponds to the background class and the remaining channels represent tumor classes. The tumor probability is computed as the aggregation of all tumor-related classes:

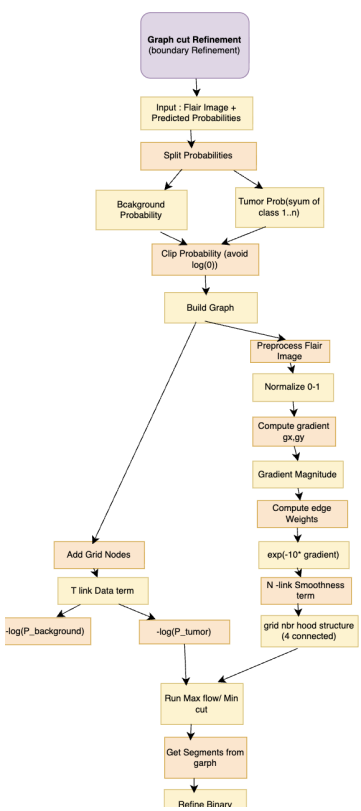


Fig. 3: Graph Cut Architecture

$$p_{\text{tumor}} = \sum_{k=1}^{K} p_k \tag{11}$$

To ensure numerical stability during logarithmic transformation, the probabilities are bounded within a small positive range:

$$p \leftarrow \text{clip}(p, \epsilon, 1) \tag{12}$$

where ϵ is a small constant. The unary (data) term is then defined using the negative log-likelihood:

$$D_p(L_p = 0) = -\log(p_{\text{background}}), \quad D_p(L_p = 1) = -\log(p_{\text{tumor}}) \tag{13}$$

This formulation ensures that pixels with higher predicted probability for a class incur lower energy when assigned that label.

The pairwise term, which enforces spatial smoothness, is derived from the intensity gradients of the input MRI image. The FLAIR modality is normalized to reduce intensity variation:

$$I' = \frac{I - \min(I)}{\max(I) - \min(I) + \epsilon} \tag{14}$$

The gradient magnitude is computed to capture local intensity changes:

$$|\nabla I| = \sqrt{\left(\frac{\partial I'}{\partial x}\right)^2 + \left(\frac{\partial I'}{\partial y}\right)^2} \tag{15}$$

Based on this, the edge weights between neighboring pixels are defined as:

$$w_{p,q} = \exp(-\beta |\nabla I|_{p,q}) \tag{16}$$

where β is a scaling parameter controlling sensitivity to intensity differences. This formulation assigns lower weights to edges with high gradients (i.e., likely boundaries), thereby preserving object boundaries during segmentation.

A grid-based graph structure is constructed where each pixel corresponds to a node, and edges are formed between neighboring pixels using a 4-connected neighborhood system. The overall energy function is expressed as:

$$E(L) = \sum_{p} D_p(L_p) + \sum_{(p,q)} w_{p,q} \cdot \mathbf{1}(L_p \neq L_q) \tag{17}$$

where $\mathbf{1}(\cdot)$ is an indicator function that penalizes label discontinuity between neighboring pixels.

The optimal labeling L^* is obtained by solving the min-cut/max-flow problem:

$$L^* = \arg\min_{L} E(L) \tag{18}$$

This optimization partitions the graph into two disjoint sets corresponding to tumor and background, minimizing the total energy.

The final refined segmentation mask is obtained by assigning labels based on the graph partition:

$$Y^* = \text{GraphCut}(P, I) \tag{19}$$

This refinement step effectively removes isolated noisy predictions, enforces region coherence, and sharpens tumor boundaries. By integrating deep learning-based probabilistic outputs with graph-based optimization, the method achieves improved segmentation accuracy and robustness, particularly in challenging regions with ambiguous intensity patterns.

E. Harvard Atlas Mapping for Precise Tumor Localization

Atlas-based brain mapping provides a standardized anatomical reference framework that enables the transformation of voxel-level predictions into clinically interpretable brain regions. In this work, the Harvard-Oxford cortical and subcortical structural atlas is employed to map segmented tumor regions onto anatomically defined brain areas, facilitating precise spatial localization and quantitative region-wise analysis.

Given a predicted tumor segmentation mask $S(x)$ obtained from the hybrid U-Net with transformer bottleneck and graph cut refinement, the first step involves aligning the patient's brain MRI to a standard anatomical space. Let $I_{patient}$ denote the input MRI and I_{atlas} represent the standard brain template. A spatial transformation function $T(\cdot)$ is estimated such that:

$$I_{normalized} = T(I_{patient}) \rightarrow I_{atlas} \tag{20}$$

This transformation typically involves affine and non-linear registration to ensure voxel-wise correspondence between patient-specific anatomy and atlas-defined regions.

The Harvard Atlas consists of a set of labeled brain regions:

$$A = \{R_1, R_2, R_3, ..., R_n\} \tag{21}$$

where each region R_i corresponds to a distinct anatomical structure (e.g., frontal lobe, temporal lobe, basal ganglia).

After registration, each voxel v in the normalized space is assigned an atlas label:

$$L(v) = \arg\max_i P(R_i|v) \tag{22}$$

This mapping enables each tumor pixel to be associated with a specific anatomical region.

Let $S(v)$ denote the binary tumor segmentation output after graph cut refinement:

$$S(v) = \begin{cases} 1, & \text{if voxel } v \text{ is classified as tumor} \\ 0, & \text{otherwise} \end{cases}$$

The tumor region mapping is defined as:

$$M(R_i) = \sum_{v \in R_i} S(v) \tag{23}$$

where $M(R_i)$ represents the tumor burden within region R_i, measured as the number of tumor-positive voxels.

F. Tumor Localization and Quantification

To provide precise localization, the probability of tumor presence in each anatomical region is computed as:

$$P(R_i|Tumor) = \frac{M(R_i)}{\sum_{j=1}^{n} M(R_j)} \tag{24}$$

This enables ranking of brain regions based on tumor involvement, allowing clinicians or decision-support systems to identify the most affected anatomical structures.

G. Non-Clinical Decision Support Interpretation

The atlas-mapped tumor distribution provides interpretable outputs by transforming voxel-level segmentation results into anatomically meaningful representations using the Harvard-Oxford brain atlas. In this framework, the primary tumor location is determined as the region with the highest probability $P(R_i \mid Tumor)$, while multi-region tumor spread is characterized by evaluating the overlap between the predicted tumor mask and atlas-defined anatomical structures. Additionally, tumor burden within each region is quantified through volume or area estimation derived from pixel-wise measurements within the corresponding anatomical boundaries.

Based on the segmented tumor mask and atlas mapping, the system generates a structured non-clinical decision support report. The tumor size and footprint analysis at the slice level indicates a total affected area of 13.66 cm^2, corresponding to 1366 pixels, and the lesion is categorized as a moderate mass. The anatomical localization identifies primary involvement of the left cerebral white matter, left cerebral cortex, and paracingulate gyrus. The locational assessment suggests a moderate risk level due to the involvement of associated functional networks, with potential impact on executive function, emotional regulation, and language or cognitive processing, particularly within the left cerebral hemisphere.

The AI-based severity estimation classifies the case as high severity, and the system suggests that immediate multidisciplinary review is advisable due to the involvement of regions associated with higher-order cognitive and executive functions. This representation enables transformation of deep learning segmentation outputs into anatomically grounded interpretations that support visualization, structured reporting, and downstream analytical modeling. Although this framework enhances interpretability and decision support, it is intended for non-clinical use and does not provide diagnostic conclusions.

H. Summary

The integration of Harvard Atlas mapping with deep segmentation and graph-cut refinement bridges the gap between pixel-level prediction and anatomically meaningful interpretation. It transforms raw segmentation outputs into region-wise tumor quantification, enhancing interpretability and enabling structured analysis of tumor distribution across brain regions.

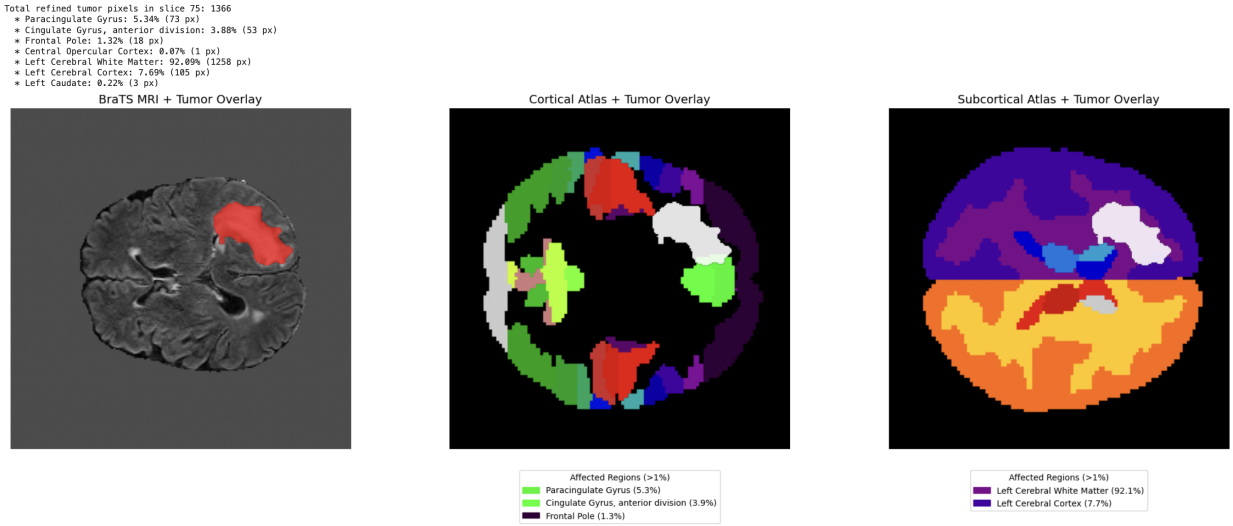


Fig. 4: Harvard Atlas-Based Tumor Localization: The segmented tumor mask is registered to the standard Harvard-Oxford brain atlas, enabling voxel-wise mapping of tumor regions onto anatomically defined brain structures for precise localization and quantification.

III. EXPERIMENT

A. Dataset

The dataset used in this study is the BraTS 2020 brain tumor MRI dataset, which contains multi-modal MRI scans along with expert-annotated segmentation masks. Each patient includes four MRI modalities: FLAIR, T1, T1ce, and T2, stored in NIfTI (.nii) format. Initially, a basic dataset loading approach is implemented where only the FLAIR modality is used. For each patient, the central slice of the 3D MRI volume is extracted along the axial plane. The corresponding segmentation mask is also extracted and resized to 128 × 128 pixels. The image intensities are normalized by scaling with the maximum pixel value, and the segmentation masks are converted into binary format (tumor vs non-tumor). To improve model performance and utilize richer information, a more advanced data generator approach is adopted. In this approach, all four modalities (FLAIR, T1, T1ce, and T2) are used. Instead of selecting a fixed slice, random slices are sampled from the range 30 to 130 during training, which increases data variability and helps the model generalize better. During validation and testing, a fixed slice index is used to maintain consistency. For each selected slice, the four modalities are stacked to form a multi-channel input (H × W × 4). Intensity normalization is applied using z-score normalization, computed only over the brain region (non-zero pixels), to reduce inter-patient intensity variations. The corresponding segmentation masks are processed by merging label 4 into label 3, ensuring consistent class representation. The final dataset used for training consists of paired multi-modal 2D slices and their corresponding segmentation masks, enabling efficient learning while preserving important tumor characteristics.

B. Evaluation Index

To evaluate the performance of the proposed brain tumor segmentation model, five standard evaluation metrics are employed: Dice Similarity Coefficient (DSC), Intersection over Union (IoU), Precision, Recall, and F1-score. These metrics quantify the agreement between the final refined segmentation mask, obtained after U-Net with transformer bottleneck and graph cut refinement, and the corresponding ground truth annotations.

The Dice Similarity Coefficient (DSC) measures the spatial overlap between the predicted tumor region and the ground truth, providing a robust indicator of segmentation accuracy. The Intersection over Union (IoU), also known as the Jaccard Index, further evaluates the overlap ratio and provides a stricter assessment of region-wise agreement. Precision quantifies the proportion of correctly identified tumor pixels among all predicted tumor pixels, reflecting the model's ability to minimize false positives. Recall, or sensitivity, measures the proportion of actual tumor pixels correctly detected by the model, indicating its effectiveness in reducing false negatives. The F1-score represents the harmonic mean of Precision and Recall, offering a balanced evaluation of segmentation performance under class imbalance conditions.

These metrics collectively provide a comprehensive evaluation of both boundary accuracy and detection reliability of the proposed hybrid segmentation framework. All metrics range from 0 to 1, where higher values indicate better segmentation performance. The formulas are defined as follows:

C. Evaluation Metrics

To evaluate the performance of the proposed brain tumor segmentation model, several standard quantitative metrics are employed, including Dice Similarity Coefficient (DSC), Intersection over Union (IoU), Precision, Recall, and F1 Score. These metrics provide complementary insights into segmentation accuracy, overlap quality, and detection reliability.

The Dice Similarity Coefficient (DSC) measures the overlap between the predicted segmentation and the ground truth. It is

defined as:

$$\text{DSC} = \frac{2TP}{2TP + FP + FN} \tag{25}$$

where TP represents true positives, FP represents false positives, and FN represents false negatives. A higher DSC indicates better agreement between predicted and ground truth regions.

The Intersection over Union (IoU) quantifies the ratio of overlap between predicted and ground truth regions relative to their union. It is expressed as:

$$\text{IoU} = \frac{TP}{TP + FP + FN} \tag{26}$$

IoU provides a stricter evaluation compared to DSC and is widely used for segmentation quality assessment.

Precision measures the proportion of correctly predicted tumor pixels among all predicted tumor pixels:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{27}$$

A high precision indicates a low false positive rate, meaning fewer normal tissues are incorrectly classified as tumor.

Recall measures the ability of the model to correctly identify all tumor pixels:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{28}$$

High recall indicates that the model successfully captures most of the tumor region with minimal false negatives.

The F1 Score represents the harmonic mean of precision and recall, providing a balanced measure between them:

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{29}$$

It is particularly useful when there is an imbalance between classes, as it ensures both false positives and false negatives are considered. In these equations, tumor pixels are considered as the positive class, while non-tumor pixels are treated as the negative class. Here, TP (True Positive) represents correctly predicted tumor pixels, FP (False Positive) represents incorrectly predicted tumor pixels, and FN (False Negative) represents tumor pixels that are missed by the model.

D. Analysis of Experimental Results

To verify the effectiveness of the proposed method, a comparison is performed with two baseline models, namely U-Net 3D and Res-UNet 3D, using standard evaluation metrics such as Dice Similarity Coefficient (DSC), Recall, and Precision across different tumor regions. The segmentation results obtained using U-Net 3D show that some tumor regions, particularly in the Whole Tumor (WT), are not fully detected, resulting in lower performance. Res-UNet 3D improves the segmentation quality by producing results that are closer to the ground truth, especially within the tumor regions, but it still falls short in capturing complete tumor boundaries.

The proposed model demonstrates strong and consistent performance across both segmentation and training stability metrics. It achieves a Dice Similarity Coefficient of 94.21% and an IoU of 86.85%, indicating high spatial agreement between predicted and ground truth tumor regions. The precision of 92.91% and recall of 94.62% reflect a balanced capability in minimizing both false positives and false negatives, resulting in an F1-score of 92.96%. Furthermore, the model attains a high accuracy of 99.38% with a low training loss of 0.0178, while validation accuracy and loss values of 99.05% and 0.0255 respectively demonstrate strong generalization performance. The small learning rate of 6.25×10^{-6} contributes to stable convergence during training.

TABLE I: Performance Evaluation of the Proposed Brain Tumor Segmentation Model

Metric	**Value**
Dice Similarity Coefficient (DSC)	94.21%
Intersection over Union (IoU)	86.85%
Precision	92.91%
Recall	94.62%
F1 Score	92.96%
Accuracy	99.38%
Loss	0.0178
Validation Accuracy	99.05%
Validation Loss	0.0255
Learning Rate	6.25×10^{-6}

TABLE II: DICE INDEX EVALUATION RESULTS OF THREE MODELS

Network	**WT**	**ET**	**TC**
U-Net 3D	0.7294	0.7427	0.7231
Res-UNet 3D	0.9224	0.8818	0.8574
Ours	0.9421	0.9351	0.9049

TABLE III: RECALL INDEX EVALUATION RESULTS OF THREE MODELS

Network	**WT**	**ET**	**TC**
U-Net 3D	0.7315	0.7254	0.7182
Res-UNet 3D	0.9042	0.8918	0.8874
Ours	0.9462	0.9329	0.9149

TABLE IV: PRECISION INDEX EVALUATION RESULTS OF THREE MODELS

Network	**WT**	**ET**	**TC**
U-Net 3D	0.7362	0.7425	0.7238
Res-UNet 3D	0.9157	0.9018	0.8975
Ours	0.9291	0.9157	0.9102

In contrast, the proposed method produces segmentation results that are more accurate and closely aligned with the ground truth for both the whole tumor region and internal tumor structures. Quantitatively, the proposed model achieves a Dice Similarity Coefficient (DSC) of 0.9421, indicating strong overlap between predicted and ground truth tumor regions. The Intersection over Union (IoU) is 0.8685, further demonstrating high spatial agreement in segmentation boundaries. In terms of detection performance, the model achieves a precision of 0.9291, reflecting a low false positive rate, and a recall of 0.9462, indicating strong sensitivity in capturing tumor regions. The F1 score of 0.9296 confirms a balanced trade-off between precision and recall.

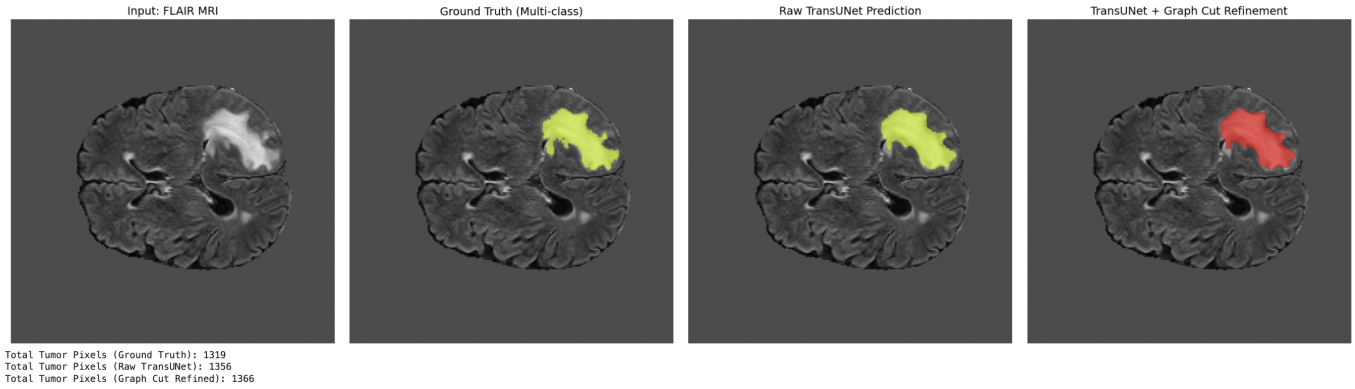


Fig. 5: Qualitative segmentation result of the proposed brain tumor segmentation model.

Overall, the proposed method consistently demonstrates superior segmentation performance across all evaluation metrics. The high DSC and IoU values indicate accurate structural alignment with ground truth annotations, while the strong precision and recall values highlight its robustness in both tumor detection and boundary delineation. These results confirm that the integration of U-Net with transformer bottleneck, followed by graph cut refinement and atlas-based interpretation, significantly enhances segmentation accuracy and reliability for brain tumor analysis.

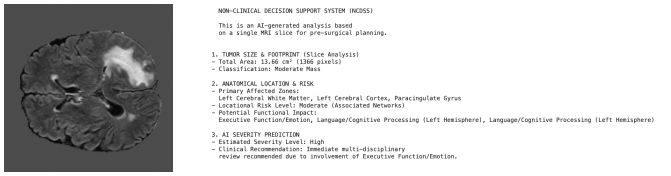
E. Non-Clinical Decision Support System Output

The proposed Non-Clinical Decision Support System is designed to transform segmentation outputs into structured and interpretable analytical reports for pre-surgical planning assistance. The system operates on a single MRI slice and integrates quantitative tumor characteristics, anatomical localization, and rule-based severity estimation to generate a comprehensive non-diagnostic assessment of brain tumor cases.

The tumor size and spatial footprint are computed directly from the predicted segmentation mask by converting pixel-wise tumor regions into physical area measurements using voxel spacing information. This enables estimation of the total tumor burden within a given MRI slice. Based on the computed area, the lesion is categorized into predefined classes such as small focal lesion, moderate mass, or large mass, providing an initial quantitative assessment of tumor extent.

For anatomical interpretation, the segmented tumor regions are mapped onto brain structures using atlas-based localization. This allows identification of primary affected zones such as cortical and subcortical regions, including white matter, cerebral cortex, and functional gyri. The locational risk is then assessed by analyzing tumor overlap with functionally significant brain regions. This enables categorization of risk levels such as low, moderate, or high depending on the involvement of critical neural networks.

In addition, the system estimates potential functional impact by associating affected anatomical regions with their corresponding neurological roles, such as executive function, emotional regulation, and language processing. This mapping provides a functional interpretation of tumor location, supporting higher-level reasoning about possible neurological



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NON-CLINICAL DECISION SUPPORT SYSTEM (NCDSS)

This is an AI-generated analysis based
on a single MRI slice for pre-surgical planning.


1. TUMOR SIZE & FOOTPRINT (Slice Analysis)
- Total Area: 13.66 cm² (1366 pixels)
- Classification: Moderate Mass

2. ANATOMICAL LOCATION & RISK
- Primary Affected Zones:
  Left Cerebral White Matter, Left Cerebral Cortex, Paracingulate Gyrus
- Locational Risk Level: Moderate (Associated Networks)
- Potential Functional Impact:
  Executive Function/Emotion, Language/Cognitive Processing (Left Hemisphere), Language/Cognitive Processing (Left Hemisphere)

3. AI SEVERITY PREDICTION
- Estimated Severity Level: High
- Clinical Recommendation: Immediate multi-disciplinary
  review recommended due to involvement of Executive Function/Emotion.
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Fig. 6: input MRI slice and the generated Non-Clinical Decision Support System (NCDSS) report output.

involvement. Finally, an AI-based severity prediction module combines tumor size and anatomical risk to generate a unified severity score. The severity estimation is formulated as a rule-based fusion mechanism that jointly considers spatial tumor burden and functional brain involvement. Specifically, larger tumor areas contribute to higher severity due to increased structural impact, while involvement of eloquent or functionally critical brain regions such as motor, sensory, or cognitive processing areas further escalates the severity level. Based on this combined assessment, cases are categorized into severity levels such as low, moderate, high, and critical.

The rule-based decision logic ensures that even moderately sized tumors are assigned higher severity when located in functionally sensitive regions, whereas smaller lesions in non-eloquent areas are assigned lower severity levels. This hierarchical fusion strategy allows the system to reflect both anatomical importance and lesion extent in a unified representation. The output includes a severity level classification along with a recommendation for multi-disciplinary review, particularly in cases where critical brain regions are involved or when the predicted tumor burden exceeds predefined thresholds.

Overall, the NCDSS converts raw segmentation outputs into structured, interpretable, and anatomically grounded reports, enabling enhanced understanding of tumor characteristics without providing direct clinical diagnosis. This transformation facilitates meaningful interpretation of deep learning predictions in a clinically aligned but non-diagnostic decision-support framework.

IV. Conclusion

In this work, a comprehensive and hybrid brain tumor segmentation and interpretation framework has been proposed by integrating a U-Net based encoder-decoder architecture

with a transformer bottleneck, followed by graph cut refinement and Harvard-Oxford atlas-based anatomical mapping. The proposed architecture effectively combines local spatial feature extraction with global contextual dependency modeling, enabling highly precise and structurally consistent tumor segmentation from MRI data. The transformer bottleneck enhances long-range spatial relationships and global context awareness, while the graph cut refinement further improves boundary regularization by enforcing spatial coherence and reducing segmentation noise, leading to anatomically smooth and visually consistent tumor masks.

The model demonstrates strong and consistent quantitative performance across multiple evaluation metrics, achieving a Dice Similarity Coefficient of 94.21%, Intersection over Union of 86.85%, precision of 92.91%, recall of 94.62%, and an F1-score of 92.96%. The overall accuracy of 99.38%, along with low training and validation loss values, further confirms the robustness and stability of the proposed framework. These results indicate that the model effectively captures both fine-grained tumor boundaries and global structural information, while maintaining a balanced trade-off between sensitivity and specificity across diverse MRI cases.

In addition to segmentation accuracy, the integration of Harvard-Oxford atlas mapping significantly enhances the interpretability and clinical relevance of the system. By transforming voxel-level predictions into anatomically grounded representations, the framework enables precise localization of tumor regions within cortical and subcortical brain structures. This region-wise mapping facilitates quantitative analysis of tumor distribution and allows the identification of functionally critical brain areas potentially affected by tumor growth, thereby improving spatial interpretability beyond pixel-level segmentation.

Furthermore, the proposed non-clinical decision support system introduces an additional interpretability layer by generating structured analytical reports derived from segmentation outputs. These reports include tumor size estimation, anatomical risk assessment, functional region involvement, and severity classification. The severity estimation mechanism integrates both spatial tumor burden and anatomical significance, providing a unified risk-oriented interpretation. This enhances the usability of the system in pre-surgical planning contexts and research-oriented decision analysis, while maintaining a non-diagnostic and supportive role.

Overall, the proposed framework not only achieves high segmentation accuracy but also bridges the gap between deep learning-based medical image analysis and anatomically interpretable decision support systems. The integration of transformer-based feature learning, graph cut refinement, and atlas-guided localization results in a robust, explainable, and scalable pipeline for brain tumor analysis. Future work will focus on extending the framework to full 3D volumetric MRI analysis, incorporating multi-modal imaging modalities such as T1, T2, and FLAIR, and integrating uncertainty quantification mechanisms to further enhance reliability, robustness, and clinical interpretability in real-world neuroimaging applications.

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